INCH-POUND
A-A-59152E
June 13, 2018

SUPERSEDING
A-A-59152D
September 23, 2010

# COMMERCIAL ITEM DESCRIPTION

TRUCK, FIRE-FIGHTING, (RESCUE) (MEDIUM RESCUE TRUCK)

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers a commercial, medium rescue truck. It has a 4x4 chassis with a diesel engine and an automatic transmission; a cab to accommodate at least three seated persons with four doors; and a utility body. The medium rescue truck is intended to stow and transport forcible-entry equipment, and rescue equipment. It is also intended to transport personnel for hazardous materials mitigation and containment. Transported personnel are also expected to perform structural, automotive, and aircraft occupant rescue, in both on- and off-road environments.

2. SALIENT CHARACTERISTICS. The medium rescue truck shall meet the requirements of National Fire Protection Association (NFPA) 1901, Standard for Automotive Fire Apparatus, (see table I) as applicable to the vehicle configuration specified.

TABLE I. Applicable NFPA 1901 requirements.

| Chapter | Title |
|---|---|
| 1 | Administration |
| 2 | Referenced Publications |
| 4 | General Requirements |
| 10 | Special Service Fire Apparatus |
| 12 | Chassis and Vehicle Components |
| 13 | Low-Voltage Electrical Systems and Warning Devices |

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: WR-AFLCMC/WNZEB, 235 Byron St., Suite 19A, Robins AFB GA 31098-1813 or emailed to SPEC99@us.af.mil/. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.dla.mil/.

FSC 4210

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25 Winches

2.1 Administration.

2.1.1 Environmental conditions.

2.1.1.1 Altitude. Unless otherwise specified (see 6.2), the vehicle shall be designed for operation at 2,000 feet above sea level.

2.1.1.2 Temperature range. The vehicle shall be capable of satisfactory storage and operation in temperatures ranging from 0 to 125 °F. The vehicle shall be equipped with a winterization system that includes an engine coolant heater and a battery heater (a blanket-type battery heater is not acceptable). The winterization system shall be powered through the electrical shoreline connection (see 2.6.2).

2.1.2 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached to the vehicle with wire ropes or chains. "Dog tag" style, beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners.

2.1.3 Roadability.

2.1.3.1 Operating terrain. The vehicle shall operate on paved and graded gravel roads and off-road (cross country) terrain.

2.1.3.2 Acceleration. The fully-loaded vehicle shall accelerate from 0 to 50 miles per hour (mph) within 35 seconds on a level, paved road.

2.1.3.3 Maximum speed. The fully-loaded vehicle shall maintain a minimum top speed of 65 mph on a level, paved surface.

2.1.3.4 Gradeability. The fully-loaded vehicle shall be able to maintain a speed of at least 5 mph while ascending any paved slope, up to and including 30 percent.

2.1.4 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

Length - 360 inches

Width - 102 inches (excluding mirrors)



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Height - 124 inches

2.1.5 Turning diameter. The fully-loaded tanker shall have a wall-to-wall turning diameter of less than three times the length of the vehicle.

2.1.6 Angles of approach and departure. The fully-loaded vehicle shall have angles of approach and departure of not less than 11°.

2.1.7 Vehicle stability. The vehicle shall comply with rollover stability criteria as defined by NFPA 1901.

2.1.8 Manuals.

2.1.8.1 Technical manuals. The overall format for the manuals may be military specification, commercial, or a combination of both. Each technical manual shall have a title page.

a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification 30 days before the first production test. Any changes or corrections noted by the procuring activity shall be corrected. Updated pages or manuals shall be submitted to the procuring activity.

b. Once approved by the procuring activity, the contractor shall pack two complete sets of technical manuals with each vehicle. An additional two complete sets shall be submitted to the procuring activity for stock.

c. Once approved by the procuring activity and a technical order (TO) number is assigned, the contractor shall pack two complete sets of both hard copy and CDs with each vehicle, and one complete set of both hard copy and CDs shall be submitted to the procuring activity for stock. The procuring activity’s address will be provided.

d. The contractor shall grant the United States Air Force (USAF) a non-exclusive, non-assignable, royalty-free U.S. Government Purpose License, to scan into CENTRA (the USAF repository), and to reproduce and distribute (either electronically or via hard medium) copies or facsimiles of manuals produced and distributed by the contractor for this CID. These rights extend to Government agencies only, and the data contained in the technical manuals is not to be sold, disclosed, or otherwise provided to any other entity or entities outside the Government. The license shall remain in effect as long as the vehicles described in the required technical manuals remain under Government control and usage. In addition, the contractor shall grant permission for the Government to place an Air Force TO number and publication date, as well as distribution, warning, handling, and destruction statements on the cover or title page of the TO.

2.1.8.1.1 Operator's manuals. The operator's manual shall include all information required for the safe and efficient operation of the vehicle, including any special attachments or auxiliary equipment. The operator's manual shall include at least the following:

a. Location and function of all controls and instruments shall be illustrated and fully described.



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b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

c. Checks and adjustments in preparation for placing the vehicle in service upon receipt from the contractor.

d. Preparation for shipment or storage.

e. Warranty information and warranty period for the complete vehicle, and for any component warranty that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers.

f. General description of the vehicle, and step-by-step instructions for the operation of the vehicle and its auxiliary equipment.

g. Description of the post-operational procedures (draining, flushing, etc.).

h. Checklists for the daily maintenance inspection and mission-readiness checks that the operator is expected to perform.

i. Procedures for towing the medium rescue truck if it becomes disabled.

j. Schedules for required preventative maintenance and required periodic maintenance.

2.1.8.1.2 Service manuals. The service manual shall identify any special tools and test equipment required, and shall cover troubleshooting and maintenance, as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index, as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Hydraulic schematic,

b. Pneumatic schematic,

c. Electrical schematic,

d. Winterization schematic,

e. Schedules for required preventative maintenance and required periodic maintenance, and

f. Location, procedure, and interval for parts of the vehicle and equipment which require lubrication.

2.1.8.1.3 Parts manuals. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers, which correspond to the reference numbers in the parts lists. All purchased parts shall be



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cross-referenced with the original manufacturer's name and part number. The parts identification manual shall provide the description, length, dimensions, and quantity of each item used per vehicle. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.1.9 Painting, plating, and corrosion control.

2.1.9.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

2.1.9.2 Finish. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint in accordance with all of the paint manufacturer's instructions and recommendations. Unless otherwise specified (see 6.2), the exterior finish color shall be Candy Apple Red, Sikkens color number FLNA3021, DuPont color number 97902U or 4737U, PPG color number 71528 (the PPG name for this color is Cardinal Red), or equal. When specified, the exterior finish color shall be Desert Sand, color number 30313 of SAE-AMS-STD-595.

a. For vehicles painted candy apple red, the cab upper body (from the bottom of the windshield) and roof shall be painted White, color number 17875 of SAE-AMS-STD-595. Compartment interiors shall have a standard commercial finish. All bright-metal and anodized parts, such as mirrors, horns, light bezels, and tread plate, shall not be painted. Roll-up compartment doors may be painted or unpainted. Front driver and passenger side doors shall include the “United States Air Force” logo in gold leaf lettering and the Air Force “Fire Emergency Services” logo at a location to be determined during the post award meeting.

b. For vehicles painted desert sand, all exterior surfaces, including all normally bright-metal and anodized parts and any interior surfaces visible with any compartment door open (but not the interior of the cab), shall be painted body color. This includes compartment shelves and mounting hardware, but does not include items mounted in the compartments. Non-metallic materials may be black or gray. Vehicles painted desert sand shall have no lettering or logo.

2.1.9.3 Horizontal reflective stripes. Horizontal reflective stripes, in accordance with NFPA 1901, shall be applied around the vehicle in an approximate plane with the headlights. Bright-metal trim or anodized parts may interrupt the reflective stripes. The pattern shall be 10-inches wide with three reflective stripes (one inch reflective, one inch body color, six inches reflective, one inch body color, and one inch reflective). The reflective stripes shall be white for vehicles painted candy apple red and desert sand.

2.1.9.4 Chevron striping. In accordance with NFPA 1901, at least 50 percent of the rear-facing vertical surfaces, visible from the rear of the apparatus, excluding any pump panel areas not covered by a door, shall be equipped with retroreflective striping in a chevron pattern, sloping downward and away from the centerline of the vehicle at an angle of 45°. 3M part number 983-17 (red) and 983-23 (florescent yellow/green) reflective diamond grade or equivalent shall be used. Each stripe in the chevron shall be a single color, alternating between red and fluorescent yellow in accordance with NFPA 1901. Each stripe shall be 6 inches (150 mm) in width in accordance with NFPA 1901 and the example provided in figure 1.



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Figure 1. Chevron stripping example.

2.1.10 Identification plate. A permanently-marked identification plate shall be securely mounted on the driver's compartment. The identification plate shall contain the following information:

a. NOMENCLATURE

b. MANUFACTURER'S MAKE AND MODEL

c. MANUFACTURER'S SERIAL NUMBER

d. USAF VEHICLE REGISTRATION NUMBER

e. NATIONAL STOCK NUMBER (NSN)

f. VEHICLE CURB WEIGHT: kg (pounds)

g. PAYLOAD, MAXIMUM: kg (pounds)

h. GROSS VEHICLE WEIGHT: kg (pounds)

i. FUEL CAPACITY AND TYPE: gal (gallons)/L (liters)



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j. DATE OF DELIVERY (month and year)

k. WARRANTY (months and km (miles))

l. CONTRACT NUMBER

m. ORDER NUMBER

n. PAINT COLOR AND NUMBER

o. LENGTH, WIDTH, AND HEIGHT OF VEHICLE IN INCHES AND CENTIMETERS

2.1.11 Delivery. All vehicles shall be driven to the receiving installation or port-of-call for OCONUS locations.

2.1.12 Warranty. The warranty period shall not begin until training has been completed at the receiving installation.

2.1.13 Air Force registration numbers. The manufacturer shall maintain a list of Air Force registration numbers and corresponding serial numbers for the purposes of identifying vehicles affected by service bulletins.

2.2 Chassis and vehicle components.

2.2.1 Capacity. The vehicle shall have a net payload capacity of 5,000 pounds of equipment, and an additional allowance for four crew members and a minimum gross vehicle weight rating (GVWR) of 29,000 pounds.

2.2.2 Engine. The vehicle shall have a diesel engine that is certified to comply with the Environmental Protection Agency (EPA) on-highway emission requirements at the time of manufacture. The engine shall run on ultra-low-sulfur, diesel fuel, unless otherwise specified (see 6.2).

2.2.2.1 Fuel filters. Primary and secondary fuel filters and a fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve, and shall be in accordance with SAE J1839. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be replaceable without loss of engine prime.

2.2.3 Exhaust system. The exhaust system outlet(s) shall be directed away from personnel accessing equipment compartments and the roof-access ladder.

2.2.3.1 Vehicle-mounted diesel-exhaust filter system. A vehicle-mounted, direct-source capture, exhaust filter system shall be supplied with the vehicle that shall prevent exposure to, and contamination from, exhaust emissions in addition to the manufacturer's after treatment device (ATD) in accordance with NFPA 1500. This system shall be installed after the engine manufacturer's ATD and before the diffuser tip in the end-tailpipe. This filter system shall work automatically whenever the vehicle exits and returns to the station. In addition, the system must have the capability to be used while on-scene, outside of the station. The diesel-exhaust removal



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system shall travel with the vehicle. The system shall not require building modifications or hanging hoses for the system to operate. The manufacturer shall install the exhaust removal system and shall not be subcontracted. The vehicle-mounted exhaust filter system shall meet all NFPA, National Institute for Occupational Safety and Health, and OSHA standards for preventing exposure to carcinogenic compounds that exist in diesel exhaust.

2.2.4 Brake system. The vehicle shall be equipped with an all-wheel, antilock brake system; the brakes shall be fully air-actuated. Brakes shall be in accordance with Code of Federal Regulations (CFR), Title 49, Sections 393.40 through 393.42(b), and 393.43 through 393.52. The braking system, complete with all necessary components, shall include:

a. Air compressor having a capacity of not less than 15 standard cubic feet per minute (scfm).

b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the reservoir tank.

c. Automatic moisture ejector on air storage reservoir.

d. Automatic slack adjusters on cam brakes, or internal self-adjusting brakes on wedge and disc brakes on all axles.

e. Spring-set parking brakes with a control located within reach of both the driver and the Officer.

f. All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of wheel rims to ensure, in case of a flat tire, that the weight of the vehicle will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

2.2.4.1 Air dryer. A replaceable cartridge, desiccant air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically-controlled heater to prevent icing of the purge valve.

2.2.4.2 Compressed-air shoreline. A flush-mounted, checked, auto-eject compressed-air shoreline connection shall be provided to maintain brake system pressure while the vehicle is not running. It shall be located on the exterior of the vehicle, either adjacent to the driver's door or within 6 inches of the left side, front corner of the body. An air chuck shall be provided as loose equipment.

2.2.5 Tires and wheels. The vehicle shall be equipped with tubeless, steel-belted, radial-ply tires with on-/off-road type tread mounted on steel wheel assemblies. Tire and wheel assemblies shall be identical at all positions. A spare tire-and-wheel assembly shall be provided; however, it is not required to be mounted on the vehicle. The spare tire(s) and rim(s) shall be inventoried, crated, and shipped as loose equipment.



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2.2.6 Fenders. Rear fenders and fender liners having tire chain clearance shall be provided. For vehicles painted desert tan, the fender flares shall be painted desert tan.

2.2.7 Chocks. Two folding wheel chocks and mounting hardware shall be provided as loose equipment. The wheel chocks shall hold the apparatus, when loaded to its GVWR or gross combination weight rating, on a hard surface with a 20-percent grade with the transmission in neutral and the parking brake released.

2.2.8 Steering. The vehicle shall be equipped with power steering.

2.2.9 Transmission. A fully-automatic transmission with a hydraulic torque converter and at least five forward speeds shall be provided. The normal driving range selector position shall provide at least four gear ratios without movement of the selector. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

2.2.10 Traction control. The vehicle shall be equipped with limited-slip, automatic-locking, or driver-controlled, air-actuated, locking differentials in the rear axle and between the axles.

2.2.11 Fuel tank. Fuel tank(s) having a minimum usable capacity in accordance with NFPA 1901 or 50 gallons, whichever is greater, shall be provided. The location of the fuel tank(s) shall protect it (them) from mechanical damage during normal use of the vehicle.

2.2.12 License-plate bracket. A lighted license-plate bracket shall be provided at the left rear of the vehicle.

2.3 Low-voltage electrical systems. The vehicle shall have a 12-volt electrical and starting system.

2.3.1 Alternator. A single alternator charging system shall be provided in accordance with NFPA 1901. The minimum continuous electrical load shall include operation of the air conditioning system.

2.3.2 Batteries. Batteries shall be of the maintenance-free type. The addition of water shall not be required during the battery's normal service life. The battery cover and vent system shall be designed to prevent electrolyte loss during service and to keep the top of the battery free from electrolyte.

2.3.2.1 Battery compartment. The batteries shall be mounted in an acid-resistant tray and shall be enclosed in a weatherproof box or compartment.

2.3.3 Battery charger or conditioner. The vehicle shall have a DC, taper-type battery charger or an automatic battery conditioner providing a minimum 15-amp output. The charger/conditioner shall be permanently mounted on the vehicle in a properly ventilated, accessible location. The charger/conditioner shall be powered from the electrical shoreline receptacle (see 2.6.2). A charging indicator shall be installed next to the receptacle. When a battery conditioner is provided, it shall monitor the battery state-of-charge and, as necessary, automatically charge or maintain the batteries without gassing, depleting fluid level, overheating, or overcharging.



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2.3.4 Warning lights. All warning lights shall use light-emitting diode (LED) elements. The warning light system, related components, and devices shall be in accordance with NFPA 1901.

2.3.4.1 Light bar. A six element, LED light bar, with both forward- and side-facing heads, shall be mounted on the cab roof. Forward-facing lenses shall be red-white-red-red-white-red, with the white lights switched off in blocking right-of-way mode. Rearward-facing lenses shall be red-amber-red-red-amber-red. The light bar shall be separately switched from the warning light panel. All vehicles shall be equipped with a steady red, front warning light as required by California state law regardless of shipping destination.

2.3.4.2 Warning light color. When specified (see 6.2), the rearward, red lights shall be replaced with amber. When specified (see 6.2), all red warning lights shall be replaced with blue.

2.3.4.3 Headlight flashing system. A high beam, alternating/flashing, headlight system shall be provided. The headlight flasher shall be separately switched from the warning light panel.

2.3.5 Audible warning devices.

2.3.5.1 Siren. The vehicle shall be equipped with an electronic siren system. The amplifier unit shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "Hi-Lo" (European) modes, and a magnetic noise-canceling microphone. The amplifier, microphone, and controls shall be within reach of the driver and right front passenger. Siren activating foot switches shall be located in front of the driver and the right front passenger. The siren speaker shall be rated at 100 watts minimum, and shall be located in a guarded position in the front bumper.

2.3.5.2 Horn. Dual, forward-facing air horns shall be installed in protected locations near the front of the vehicle. Air-horn-activating foot switches shall be located in front of the driver and the right front passenger.

2.3.6 Work lighting.

2.3.6.1 Cab interior lights. Cab interior-light levels shall be a minimum of 2 foot-candles.

2.3.6.2 Compartment lights. White lighting, sufficient to provide an average minimum illumination of 2 foot-candles, shall be provided in each compartment greater than 4 cubic feet and having an opening greater than 144 square inches. Where a shelf is provided, this illumination shall be provided both above and below the shelf. Lights shall automatically illuminate only when the respective doors are opened.

2.3.6.3 Ladder, step, and area lights. Non-glare, white lighting shall be provided at ladders and access steps where personnel work or climb during night operations. These area lights shall be controlled with switches on the cab instrument panel and near the light sources.

2.3.7 Scene lights. A total of eight 12-volt, LED, high-mounted floodlights shall be provided to illuminate the work areas around the vehicle: two on each side, two at the rear, and two in the front. Individual switches shall be located in the work areas and on the instrument panel. A switch shall be provided to control the lights on the left side of the vehicle, a second switch shall be provided to control the lights on the right side of the vehicle, and a third switch shall be provided



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to control the lights at the rear of the vehicle. A separate switch for the front lights shall be located on the instrument panel only. Three-way switches shall be used.

2.3.8 Radio circuit. The vehicle shall have two separate 30-amp circuits with breakers, and at least 6-feet of wire routed to a space provided adjacent to the driver and turret operator for purchaser-provided radios and other electrical equipment. The wiring shall be tagged indicating its purpose.

2.3.9 Radio antennas. Two antennae connections with coaxial cable and waterproof caps shall be provided for two, 2-way radios. The mounts shall be located on the cab roof to the rear of the front seats and will be clear of any obstructions. The cables shall be routed to the center, floor-mounting console; length of the cables shall be determined at post award.

2.4 Driving and crew areas.

2.4.1 Cab. The vehicle shall have a tilting or non-tilting cab with four doors that can seat at least four persons. The crew cab area shall be fitted with a lockable medical-supply cabinet in between the two rear crew-member seats. The height of the cabinet shall extend from the cab floor to the ceiling of the cab. The cabinet shall be provided with two adjustable shelves. The width and depth of the cabinet shall be the maximum possible without hindering the movement and comfort of the firefighters with full gear in the cab. Steps and handrails shall be provided for all crew doors, and at least one grab handle shall be provided for each crew member located inside the cab for use while the vehicle is in motion. The area under the cab doors shall be covered with aluminum tread plate. A vehicle data recorder shall be provided in accordance with NFPA 1901. Provisions must be made for fire-helmet storage for all seated positions in accordance with of NFPA 1901. The cab shall have a console for radio mounting, be equipped with adjustable work lights, and be located between the driver and front passenger seats.

2.4.1.1 Seats. The vehicle shall have seats in accordance with NFPA 1901. The driver's seat shall be of an adjustable air-suspension design. The crew-member seats shall be of a non-suspension design. Each crew-member seat (excluding the drive position) shall have a backrest, brackets, and pad covers designed to store a one-hour capacity self-contained breathing apparatus (SCBA). Each seat shall be provided with a type 2 seat-belt assembly, i.e., 3-point restraint, in accordance with 49 CFR 571.209. All seats shall be equipped with a seat-belt sensor system in compliance with NFPA 1901. All seat belts shall be red in color.

2.4.1.2 Cab interior sound level. The maximum sound level at any seat location shall not exceed 90 dBa without any warning devices in operation, as measured in accordance with 49 CFR 393.94(c), "Interior noise levels in power units, test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth-surface road.

2.4.1.3 Windshield and windows. The windshield and windows shall be of tinted safety glass.

2.4.1.4 Heated mirrors. Remotely-controlled, powered, heated, rearview mirrors shall be installed outside and on each side of the cab. Mirrors shall be mounted on fold-back brackets.

2.4.1.5 Climate control system. The offeror/contractor's standard heater/defroster system shall be provided. In 100 °F ambient temperature with 50 percent relative humidity and at maximum



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compressor speed, the air conditioning system shall cool the fully-occupied cab to 75 °F within 30 minutes.

2.4.1.6 Instruments and controls. Gauges shall be provided for oil pressure, coolant temperature, and automatic transmission fluid temperature. All switches and/or controls that activate systems or system components shall be labeled as to their function. In addition to the instruments and controls required by NFPA 1901, the following shall be provided within convenient reach of the seated driver:

a. Master warning-light control switch,

b. Work light switch(es), and

c. Compartment "Door Open" warning light and intermittent alarm that sounds when a compartment door is open and the parking brakes are released or the transmission is in any position other than neutral.

2.4.2 Back-up camera. A rear-vision camera and in-cab monitor shall be provided to aid the driver in safely backing up the vehicle. The monitor shall be located to the right side of the driver. A switch shall be provided to allow the driver to manually activate the back-up camera from within the cab. The back-up camera shall also be switched "on" automatically whenever the vehicle is in the reverse mode of operation.

2.5 Body, compartments, and equipment mounting.

2.5.1 Body. The vehicle shall have a body constructed entirely of non-corroding materials. The body shall be at least as strong as if it were constructed of 0.125-inch thick aluminum. A copy of the fire apparatus manufacturer's warranty shall be included with the bid. The warranty shall state that the body shall be free of structural failures caused by defective design or workmanship for a warranty period of 15 years from the date the new vehicle is first delivered or 100,000 miles, whichever occurs first. The warranty shall also state that defective parts under the warranty shall be repaired or replaced without charge to the original purchaser.

2.5.2 Compartments. The utility body shall have side compartments, and may have rear compartments, with a minimum of 450 cubic feet of enclosed storage space.

2.5.2.1 Compartment doors. Compartments shall have clear, anodized aluminum, counterbalanced, non-locking, roll-up doors and will be provided with an aluminum tray/guard that protects the roll-up door from damage. Guard shall not be provided on medical cabinets located inside of the cab. Door latch handles shall be full-width bar type. Door straps shall be provided to assist in closing the compartment doors when the rolled-up door height exceeds 6 feet above the ground.

2.5.2.2 Scuff plates. Replaceable scuff plates shall be provided to prevent body damage from sliding equipment in and out of the compartments. The scuff plates shall be attached in such a manner that does not allow them to become loose during normal equipment use.



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2.5.2.3 Drip rails. Drip rails shall be provided over each compartment door. If the drip rails are not integral with the body, they shall be of anodized, extruded aluminum, and shall have a bright finish for vehicles painted candy apple red.

2.5.2.4 Shelves. An adjustable and removable compartment shelf shall be provided for every 18 inches of each vertical storage-compartment door opening. Shelving adjustments shall require no more than common hand tools, and shall not require disassembly of fasteners. Shelves shall support a minimum of 500 pounds without permanent deformation. Each shelf shall be accessible to crew members standing on the ground. A pull-out and tip-down configuration shall be used for shelving that is over 54 inches from the ground when the compartment is more than 18 inches deep. Each shelf shall have ¼-inch diameter drain holes located in each corner of the shelves so as to allow for drainage of any water from the stowed equipment.

2.5.2.5 Drainage mats. Each compartment floor and shelf shall be covered with a removable mat designed to allow for drainage of any water from the stowed equipment.

2.5.2.6 Pass-through compartments with roll-out trays. Two pass-through compartments with roll-out trays shall be provided. The length of each roll-out tray shall be the full interior width of the utility body and shall extend 4 feet from either side of the utility body. Each roll-out tray shall support a 500-pound load without permanent deformation. The roll-out trays shall be adjustable for vertical height in 18-inch increments. Slide-out adjustments shall be accomplished without the use of tools. Automatic latches shall secure the stowed trays and shall prevent over-travel when fully extended. One roll-out tray shall support a customer-provided and -mounted, diesel-engine-powered, rescue tool kit.

2.5.3 Ladder storage compartment. The vehicle shall have a storage compartment within the utility body for a 14-foot long, 25-inch wide, two-section extension ladder. The storage compartment shall be located at the rear of the vehicle and shall permit removal of the ladder by a person standing on the ground.

2.5.4 SCBA storage rack. A rack for storage of eight SCBA, one-hour cylinders shall be installed in a compartment. The top of the rack shall not be higher than 66 inches above the ground. The tubes shall be designed to a length and diameter of sufficient size to accommodate any commercially-available, one-hour SCBA cylinder.

2.5.5 Ladder, handrails, and walkways. The utility body roof shall be covered with tread plate which shall support the weight of a crew member without permanent deformation. The utility body roof shall have an 8-inch high, cargo tie-down rail around the perimeter. The utility body shall have a roof access ladder with non-slip handrails and rungs. All stepping, standing, and walking surfaces, to include ladders, shall be in accordance with NFPA 1901. Handrails shall be provided in accordance NFPA 1901.

2.6 Line-voltage electrical system.

2.6.1 Auxiliary generator. A 15-kilowatt (KW), continuous rating, 120/240-volt, 60-hertz generator shall be provided. The generator shall be activated from the driver's position and shall have a monitor light to indicate operation. A governor shall regulate engine speed to match the generator output to the connected load. Gauges shall monitor the operation of the generator system



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and indicate the connected load. Access for maintenance shall be provided above the generator. Individual circuit breaker tripping or failure shall not affect operation of other active circuits. The generator shall interface with the Idle Reduction Technology system.

2.6.2 Electrical shoreline connection. The battery charger/conditioner shall be powered from a covered, three-wire, straight-blade, polarized, insulated, labeled, recessed, 120-volt, NEMA 5-30P male-plug inlet, in a weatherproof AC, auto-eject receptacle. It shall be located on the exterior of the vehicle, either adjacent to the driver's door or within 6 inches of the left side, front corner of the body. The shoreline connection's location shall not pose a tripping hazard to pedestrians walking by the vehicle when the shore line is connected. A weatherproof charge meter shall be installed next to the receptacle. A 50-foot long, three-wire, 30-amp rated, 120-volt, AC power cable, with straight blade (non-twist-lock), NEMA 5-30R and 5-30P style connectors, in accordance with NEMA WD-6, shall be provided. When equipped with a winterization system, the winterization system shall have a labeled activation switch with a pilot light. When specified (see 6.2), the receptacle(s) and power cable(s) shall be for 220 volts in lieu of 110 volts.

2.6.3 Receptacles. One duplex, NEMA 5-20R, 120-volt, 20-amp receptacle shall be provided at each corner of the body. Directly adjacent to the duplex receptacle, a 120-volt, 20-amp, L5-20R receptacle shall be provided. The receptacles shall be powered by the auxiliary generator (see 2.6.1). Two 120-volt AC, 20-amp, electrical receptacles shall be provided, one at each corner of the body. Each receptacle shall have weatherproof cover(s) and shall provide one L5-20R and one duplex NEMA 5-20R receptacles. The receptacles shall be powered by the auxiliary generator (see 2.6.1). See figure 2. Specific locations will be annotated to design drawings at the post-award meeting.



Figure 2. Receptacles.

2.6.4 Cable reel. The vehicle shall be equipped with an electrical cable reel mounted in the right rear compartment. The reel shall be equipped with 200 feet of 20-amp, 600-volt, 90 °C, insulated electrical cable. The electrical cable shall be equipped with a rubber-ball stop to prevent cable pull through during rewinding operations. A four-way roller guide shall be provided for the cable reel to prevent chafing of cable insulation. The cable reel shall have an electric rewind motor with provisions for manual rewind in the event of motor failure. The manual rewind handle shall be securely stored near the cable reel. A portable, lighted, weatherproof, GFCI receptacle box, with built-in circuit breakers, shall be provided to the cable end. The box shall, at a minimum, provide two NEMA L5-20R and two duplex NEMA 5-20R receptacles. The cable reel shall be powered by the auxiliary generator (see 2.6.1).



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2.6.5 Light mast. The vehicle shall be equipped with an elevating light mast with two 1,500-watt, "instant-on" halogen lights. The light mast shall be of a free-standing design, shall not require outriggers or guy wires, and shall survive 100-mph winds with no damage. The light mast shall extend a minimum of 18 feet above the ground. It shall stop at any vertical height between its maximum and minimum height. The light head shall rotate at least 350° (it is not necessary for the head to rotate when it is in the stored position). Electrical power shall be provided by the auxiliary generator (see 2.6.1). The light mast shall include a remote control on a 25-foot cord to provide pan-and-tilt and rotation of the lights. The remote control cord for the light mast shall be a detachable, tether cord. The port for the remote control cord shall be located on the exterior driver's side of the vehicle. This port shall be designed for external installation and have a weatherproof cover. A bracket for storage of the remote control shall be provided. The bracket shall be located in a utility body compartment in close proximity to the external port. The lights and light mast shall be operable by a person standing on the ground. The light system shall be interlocked to operate only while the engine is running. A master switch for controlling the activation of the "stand-up" mode of the light mast shall be provided. This switch shall have a light to indicate when the circuit is active. The switch shall be located inside the cab in an area easily accessible to the driver. Pneumatic controls, if provided, shall include a permanently-set air regulator, air control valves, and a system air pressure gauge. A red warning light shall flash continuously in front of the driver while the light mast is extended or not firmly set in its support cradle. The vehicle horn shall blow continuously if the transmission is placed in gear, or the parking brake is released while the mast is extended or not set firmly in its support cradle. The light mast system shall be in accordance with NFPA 1901.

2.6.6 Idle reduction technology (IRT). An IRT system shall be supplied with the apparatus that shall shut down the main chassis, diesel engine during operations not requiring it. This system shall be automated and shall not require intervention from the vehicle operator. The chassis voltage system shall be protected against extreme drain of the battery bank. If the vehicle's voltage drops too low, the automatic engine controls shall start the chassis diesel engine to provide a charge. The IRT system shall power an in-cab, climate control system with a minimum of 13,500 BTU A/C evaporator, 300 CFM air flow, and 5,600 BTU heating element. The IRT system shall be available when the chassis air brake is set. The system shall interface with the chassis batteries and an auto-eject, electric shoreline connection. Easy to access control panel(s) and fuse boxes shall be installed on the vehicle. The fuel supply for the IRT shall be the chassis fuel tank. At a minimum, the following systems shall be operable when the IRT is engaged: in-cab climate control system, radios, Department of Transportation truck lighting, NFPA required emergency lights, step lights, and compartment lights.

2.7 Air systems.

2.7.1 Air hose reel. An air hose reel shall be mounted in the rear compartment near the center of the vehicle. The hose reel shall be equipped with 150 feet of 0.50-inch inside diameter hose line; a 3/8-inch National Pipe Taper (NPT) fitting; and a female-style quick disconnect shall be connected to the end of the hose line. The hose line shall be equipped with rubber-ball stops to prevent hose pull through on roller guides during rewinding operations. A four-way roller guide shall be provided for the hose reel to prevent hose chafing and kinking. The hose reel shall have an electric rewind motor with provisions for manual rewind in the event of motor failure. The manual rewind handle shall be securely stored near the hose reel. A pressure-protected air supply



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from the chassis air system shall be connected to the hose reel. The air supply lines shall be routed with minimum bends and located or guarded from damage by the carried equipment.

2.8 Winch. A winch with at least 12,000-pound pull shall be installed, recessed behind the front bumper. The winch shall be electric or hydraulic powered, and shall have one or more forward and reverse speeds of not less than 15 feet per minute. The winch shall be equipped with a minimum 125 feet of 3/8-inch galvanized aircraft cable with a 36-inch end chain and hook. The winch shall include a four-way cable guide. A 25-foot minimum, remote control cable shall be provided for operation of the winch. If an extended bumper is used, a cover fabricated of tread plate shall be installed over the winch and the space between the cab and bumper.

2.9 Trailer towing package. The vehicle shall be equipped with a square, sleeve-type receiver used to tow trailers equipped with a ball tow hitch and other ancillary firefighting equipment. The tow hitch and receiver shall be rated at the maximum inherent towing capability of the vehicle such that safety and reliability of the vehicle are not degraded. The trailer hitch shall comply with NFPA 1901. The maximum towing capacity and tongue weight capacity of the vehicle shall be clearly and visibly posted, both near the tow hitch receiver and inside the cab of the vehicle. The manufacturer shall describe any hazards, risks, maximum suggested speed limits or restrictions, and special use requirements during the solicitation process. This information shall also be listed near the tow hitch receiver, inside of the cab of the vehicle, as well as in the operator's manual. A brake controller shall be provided for use with a trailer that has an electric braking system. Gladhands shall be provided for use with a trailer that has an air braking system in accordance with SAE J318.

2.10 Safety-chain attachment points. The vehicle shall be equipped with two safety-chain attachment points, one located on each side of the receiver sleeve, symmetrical about the receiver sleeve. The vertical centerline of the safety-chain attachment points shall be located at a parallel distance of not greater than 6 inches to the left and right of the vehicle centerline. Safety chains shall comply with NFPA 1901, and shall have the ability for standard safety-chain hooks to be attached. The inside radius of the safety-chain hook is between 1 inch and 3 inches.

2.11 Workmanship. The vehicle, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts; tightness of fasteners; etc. Wire ties may be used for bundling vehicle wiring, but not for support of wire bundles. Support must be provided by insulated metal clamps. The vehicle shall be thoroughly cleaned of all foreign matter.

2.12 Human factors engineering. All system operations, servicing, and maintenance functions shall be configured to be accomplished by a range of personnel from a 5$^{th}$-percentile female to a 95$^{th}$-percentile male.

2.13 Training. A minimum of two days of training shall be provided at the receiving installation following the delivery of the fire vehicle. The training shall consist of driver/operator training and informational overview training for vehicle fleet-maintenance personnel.



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3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs. The offeror/contractor is encouraged to use recovered materials or biobased products to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403. However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100 percent of EPA-designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meeting reasonable performance standards; or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

4.2 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the manufacturer to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID. The manufacturer shall provide an itemized technical proposal that describes how the proposed model complies with each salient characteristic of this CID; a paragraph-by-paragraph response to the salient characteristics section of this CID shall be provided**.** The response to each paragraph shall clearly state “Fully Comply” or “Bid with Exception” in addition to other data the contractor wishes to submit. The proposal shall be provided with the pricing submission. Failure to provide this information may deem a vendor as non-responsive and their proposal may be rejected. The manufacturer shall provide two copies of their commercial descriptive catalogs with their offer as supporting reference to the itemized technical proposal. The manufacturer shall identify all modifications made to their commercial model in order to comply with the requirements herein.

4.3 Inspection requirements.

4.3.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision-type, calibrated at proper intervals to ensure laboratory accuracy.



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4.3.2 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

j. Evidence of corrosion.

4.3.3 Detailed inspection requirements.

4.3.3.1 Examination of product. Each vehicle shall be examined to verify compliance with the salient characteristics herein. A contractor-generated checklist that identifies each relevant requirement and the inspection results shall be used. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each vehicle function shall be verified. Each production vehicle shall be inspected to a reduced version of the checklist that has been approved by the procuring activity.

4.3.3.2 Road tests. The following tests shall be performed in accordance with NFPA 1901.

4.3.3.2.1 Maximum speed and acceleration test. The vehicle shall be tested to demonstrate compliance with 2.1.3.2 and 2.1.3.3. For the first production unit, a time-distance recorder shall be used to record data for this test.

4.3.3.2.2 Gradeability test. The first production vehicle shall be tested to demonstrate compliance with 2.1.3.4.



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4.3.3.2.3 Service brake system test. The vehicle shall be tested in accordance with NFPA 1901. For the first production unit, a time-distance recorder shall be used to record data for this test.

4.3.3.2.4 Turning diameter test. The fully-loaded, first production vehicle shall be tested in accordance with SAE J695 to demonstrate compliance with 2.1.5.

4.3.3.2.5 Roadability test. The fully-loaded, first production vehicle shall be driven over 10 miles of paved and 10 miles of off-road terrains. All loads shall be removed and all structure and surfaces shall be visibly inspected for failure or permanent deformation.

4.3.3.3 Low voltage electrical system performance tests. The vehicle shall be tested in accordance with NFPA 1901.

4.3.3.4 Cab interior sound level test. The cab interior sound levels of the first production vehicle shall be measured in accordance with 49 CFR 393.94(c), "Interior noise levels in power units, test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

4.3.3.5 Line voltage electrical system testing. The vehicle shall be tested in accordance with NFPA 1901.

5. PACKAGING.

5.1 Preservation and packaging. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Government documents. Copies of specifications, standards, and handbooks are available from DLA Document Services, 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094, or online at https://assist.dla.mil.

6.1.2 Code of Federal Regulations (CFR). CFRs may be obtained from the Superintendent of Documents, U.S. Government Printing Office, 732 North Capitol Street, NW, Washington, DC 20401-0001, or online at http://www.ecfr.gov/cgi-bin/ECFR?page=browse.

6.1.3 Society of Automotive Engineers (SAE) standards. Copies of SAE standards may be obtained from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at http://www.sae.org/.

6.1.4 National Fire Protection Association (NFPA). Documents may be obtained from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471, or online at http://www.nfpa.org/codes-and-standards.



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6.1.5 Federal Acquisition Regulations (FAR). FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

6.1.6 National Electrical Manufacturers Association (NEMA) standards. Copies of NEMA standards may be obtained from National Electrical Manufacturers Association, 1300 North 17th Street, Suite 900, Arlington, VA 22209, or online at https://www.nema.org/Standards/pages/default.aspx.

6.1.7 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be obtained from Occupational Safety and Health Association, 200 Constitution Ave., NW, Washington, D.C. 20210, or online at http://www.osha.gov/.

6.1.8 Environmental Protection Agency (EPA). Copies of these documents may be obtained from Environmental Protection Agency, 1200 Pennsylvania Avenue, N.W., Washington, DC 20004, or online at http://www.epa.gov/.

6.2 Ordering data. The contract or order should specify the following:

a. Altitude for which vehicle operation is to be designed, if greater than 2,000 feet above sea level (see 2.1.1.1).

b. Finish color required (desert sand in place of candy apple red) (see 2.1.8.2).

c. If regular diesel is required in lieu of ultra-low-sulfur diesel (see 2.2.2).

d. Emergency warning light color required (amber or blue in place of red) (see 2.3.4.2) (NOTE: applies to USAFE only).

e. Electrical shoreline voltage, 220 volts instead of 110 volts (see 2.9.1) (NOTE: applies to some USAFE and Southwest Asia (CENTAF) locations only).

6.3 Key words.

entry
forcible
hazardous
stow
transport



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